

SPL (Splunk Search Processing Language) Cheat Sheet

58 core SPL commands, operators, and functions with descriptions and examples.

Command	Category	Description	Example
search	<i>Filtering</i>	The default/implicit command; filters raw events.	index=main sourcetype=access_combined status=200
where	<i>Filtering</i>	Filters using an eval-style boolean expression.	where status==200
field=value	<i>Comparison</i>	Equals, used directly in search (not where).	index=main Image="cmd.exe"
field!=value	<i>Comparison</i>	Not equal, used directly in search.	index=main EventID!=4688
==	<i>Comparison</i>	Equals, used inside where/eval.	where EventID==4688
!=	<i>Comparison</i>	Not equal, used inside where/eval.	where EventID!=4688
NOT	<i>Logical</i>	Excludes matching events.	error NOT warning
AND / OR	<i>Logical</i>	Combine search terms (AND is implicit between terms).	status=200 AND method=GET
wildcard (*)	<i>Filtering</i>	Matches any characters within a search term.	index=main Image="*powershell.exe"
like()	<i>Comparison</i>	SQL-style wildcard match, used inside where/eval (% and _ wildcards).	where like(Image, "%cmd.exe%")
cidrmatch()	<i>Network</i>	Tests whether an IP falls within a CIDR range.	where cidrmatch("10.0.0.0/8", src_ip)
sort	<i>Sorting</i>	Sorts results; prefix with - for descending.	sort -_time
dedup	<i>Deduplication</i>	Keeps only the first event per unique value(s).	dedup Image
table	<i>Column selection</i>	Displays only the specified columns, in order.	table _time, Computer, Image
fields	<i>Column selection</i>	Includes or excludes fields (use - to exclude).	fields - ProcessGuid
rename	<i>Column selection</i>	Renames a field.	rename Computer as Host
eval	<i>Column creation</i>	Creates or overwrites a field using an expression.	eval Hour=strftime(_time,"%H")
stats	<i>Aggregation</i>	Groups events and applies aggregate functions.	stats count by Image
count	<i>Aggregation function</i>	Counts events per group.	stats count by EventID
dc()	<i>Aggregation function</i>	Counts distinct values per group.	stats dc(Computer) by Image

Command	Category	Description	Example
sum()	<i>Aggregation function</i>	Sums a numeric field per group.	stats sum(bytes) by src_ip
avg()	<i>Aggregation function</i>	Averages a numeric field per group.	stats avg(duration) by user
min() / max()	<i>Aggregation function</i>	Smallest / largest value per group.	stats max(_time) by Computer
values() / list()	<i>Aggregation function</i>	Collects unique / all values into a multivalue field.	stats values(Image) by Computer
bin	<i>Bucketing</i>	Rounds a field (often time) into fixed buckets.	bin _time span=1h
top	<i>Sorting + limiting</i>	Returns the most common values of a field.	top limit=10 Image
rare	<i>Sorting + limiting</i>	Returns the least common values of a field (opposite of top).	rare Image
head	<i>Row limiting</i>	Returns the first N results.	head 20
tail	<i>Row limiting</i>	Returns the last N results.	tail 20
join	<i>Combining searches</i>	Joins results with another search on a shared field.	join type=inner ProcessGuid [search index=main EventID=1]
append	<i>Combining searches</i>	Appends the results of a second search.	append [search index=main EventID=3]
multisearch	<i>Combining searches</i>	Runs multiple searches together, interleaved by time.	multisearch [search index=a] [search index=b]
lookup	<i>Enrichment</i>	Enriches events with fields from a lookup table.	lookup registry_persistence_lookup RegistryPath AS TargetObject OUTPUT MITRE_ID
inputlookup	<i>Enrichment</i>	Loads a lookup table directly as search results.	inputlookup registry_persistence_lookup.csv
transaction	<i>Event grouping</i>	Groups related events into a single transaction.	transaction ProcessGuid maxspan=5m
streamstats	<i>Running stats</i>	Computes stats per-event as a running total (no collapse).	streamstats count as EventNum by Computer
eventstats	<i>Aggregation (non-collapsing)</i>	Adds aggregate stats to every event without grouping rows away.	eventstats avg(bytes) as AvgBytes by src_ip
timechart	<i>Aggregation + visualization</i>	Aggregates over time buckets for charting.	timechart span=1h count by Image
chart	<i>Aggregation (tabular)</i>	Aggregates into a chart-ready table (rows x columns).	chart count by Image, EventID
regex	<i>Filtering</i>	Filters events using a regular expression.	regex ServiceName="^[A-Za-z]{4}\$"

Command	Category	Description	Example
rex	<i>Field extraction</i>	Extracts a new field from another field using regex.	rex field=CommandLine "-Uri (?<Url>\S+)"
erex	<i>Field extraction</i>	Generates a regex automatically from example values.	erex Url examples="http://10.0.2.6/dghelper.dll"
mvexpand	<i>Array expansion</i>	Expands a multivalue field into separate events.	mvexpand Image
makemv	<i>Array creation</i>	Converts a delimited field into a multivalue field.	makemv delim="," Tags
spath	<i>Extraction</i>	Extracts fields from JSON or XML data.	spath input=_raw path=user.name output=UserName
convert	<i>Type conversion</i>	Converts field types (e.g. string to number).	convert num(bytes)
tonumber() / tostring()	<i>Type conversion</i>	Eval functions to convert value types.	eval PortNum=tonumber(SourcePort)
if()	<i>Conditional logic</i>	Inline if/else for a single condition.	eval IsAdmin=if(User=="SYSTEM", "Yes", "No")
case()	<i>Conditional logic</i>	Multi-branch conditional expression.	eval Severity=case(Level==1, "Critical", Level= =2, "High", 1==1, "Low")
match()	<i>Regex function</i>	Tests a field against a regex, returns true/false.	eval IsExe=if(match(Image, "\.exe\$"), "Yes", "No")
strftime() / strptime()	<i>Date/time</i>	Formats or parses a timestamp.	eval Formatted=strftime(_time, "%Y-%m-%d")
relative_time() / now()	<i>Date/time</i>	Computes a relative or current time value.	where _time > relative_time(now(), "-7d")
earliest / latest	<i>Time range</i>	Limits the search time window.	index=main earliest=-24h latest=now
subsearch []	<i>Nested search</i>	Uses the results of one search as input to another.	index=main [search index=main EventID=13 fields ProcessGuid]
fillnull	<i>Data cleanup</i>	Replaces null/missing values with a default.	fillnull value=0
coalesce()	<i>Data cleanup</i>	Returns the first non-null value from a list of fields.	eval Host=coalesce(Computer, host)
iplocation	<i>Enrichment</i>	Adds geographic fields based on an IP address.	iplocation src_ip
outputcsv / inputcsv	<i>Export/import</i>	Writes results to, or reads results from, a CSV file.	outputcsv results.csv

Note: SPL is used in Splunk Enterprise and Splunk Cloud. Field names in examples (Image, CommandLine, Computer, src_ip) are generic Sysmon/CIM-style names — adjust to match your actual index and sourcetype schema.